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SURFACE VEHICLE STANDARD

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On-Board System Requirements for
V2V Safety Communications by
Non-Light-Duty Vehicles and Motorcycles

RATIONALE

This SAE Standard specifies the system requirements for vehicle-to-vehicle (V2V) safety system for FHWA vehicle classes 1 (motorcycles) and 4 through 13 (non-light-duty), including functional requirements and performance requirements. The system can transmit and receive the SAE J2735-defined Basic Safety Message (BSM) over a wireless communications interface; the communications interface itself is outside the scope of this document. This document provides the specifications necessary to build interoperable systems that support V2V applications for non-light-duty vehicles, including motorcycles, which rely on the exchange of BSMs. The document covers vehicle classes not addressed in SAE J2945/1 and SAE J3161/1.

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1. SCOPE

This SAE Standard specifies the system requirements for vehicle-to-vehicle (V2V) safety system for FHWA vehicle classes 1 (motorcycles and 4 through 13 (non-light-duty), including functional requirements and performance requirements. The system can transmit and receive the SAE J2735-defined BSM over a wireless communications interface; the communications interface itself is outside the scope of this document. This document provides the specifications necessary to build interoperable systems that support V2V safety applications for non-lightweight vehicles, as well as motorcycles which rely on the exchange of BSMs. The document covers vehicle classes not addressed in SAE J2945/1 and SAE J3161/1.

2. REFERENCES

The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the latest issue of SAE publications shall apply.

2.1 Applicable Documents

2.1.1 SAE Publications

Available from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, Tel: 877-606-7323 (inside USA and Canada) or +1 724-776-4970 (outside USA), www.sae.org.

SAE J2735 V2X Communications Message Set Dictionary

SAE J2945 Dedicated Short Range Communication (DSRC) Systems Engineering Process Guidance for SAE J2945/X Documents and Comment Design Concepts™

SAE J2945/1 On-Board System Requirements for V2V Safety Communications

SAE J3161 LTE Vehicle-to-Everything (LTE-V2X) Deployment Profiles and Radio Parameters for Single Radio Channel Multi-Service Coexistence

SAE J3161/1 On-Board System Requirements for LTE-V2X V2V Safety Communications

2.1.2 IEEE Publications

Available from IEEE Operations Center, 445 and 501 Hoes Lane, Piscataway, NJ 08854-4141, Tel: 732-981-0060, www.ieee.org.

IEEE Std 1609.2 IEEE Standard for Wireless Access in Vehicular Environments (WAVE) - Security Services for Applications and Management Messages

2.2 FHWA Publications

Available from Federal Highway Administration, 1200 New Jersey Ave, SE, Washington, DC 20590, Tel: 202-366-4000, https://www.fhwa.dot.gov/.

FHWA Traffic Monitoring Guide, Appendix B - Vehicle Types

3. DEFINITIONS

For the purposes of this standard, the definitions, abbreviations, and acronyms found in SAE J2735 and SAE J2945/1 also apply. In the case of conflict, the following additional definitions, abbreviations, and acronyms supersede those found in SAE J2735. The definitions in SAE J2945 are incorporated by reference.

3.1 ROLL ANGLE

Roll angle is defined as the angle a motorcycle's body makes around the longitudinal axis in the direction of travel without considering the position of the rider.

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3.2 HARD BRAKING

A class 7 to 13 or a class 4 vehicle is decelerating at a level greater than 0.2 g. For all other classes, the level is greater than 0.4 g.

4. ABBREVIATIONS

The additional abbreviations and acronyms cited below are terms related to this standard (and of the other companion volumes and guides), unless specifically cited otherwise.

ABS	Antilock Brake System
ARC-IT	Architecture Reference for Cooperative and Intelligent Transportation
ASN.1	Abstract Syntax Notation One
BSM	Basic Safety Message
BSW	Blind Spot Warning
CAN	Controller Area Network
CFR	Code of Federal Regulation
CLW	Control Loss Warning
CVRIA	Connected Vehicle Reference Implementation Architecture
DE	Data Element
DF	Data Frame
DNS	Domain Name Services
DOT	Department of Transportation
DSRC	Dedicated Short Range Communications
DVI	Driver Vehicle Interface
ECU	Electronic Control Unit
FCW	Forward Collision Warning
FHWA	Federal Highway Administration
FR	Functional Requirement
GNSS	Global Navigation Satellite System
HV	Host Vehicle
ID	Identifier
IEEE	Institute of Electrical and Electronics Engineers
IT	Intelligent Transportation
IMA	Intersection Movement Assist

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LCW	Lane Change Warning
LTE-V2X	LTE Vehicle-to-Everything
MAC	Medium Access Control
MD	Model Deployment
MHz	Megahertz
MIB	Management Information Base
NRTM	Needs to Requirements Traceability Matrix
OBE	On-Board Equipment
PICS	Protocol Implementation Conformance Statement
PPS	Pulse per Second
PSID	Provider Service ID
RF	Radio Frequency
RV	Remote Vehicle
SCMS	Security Credential Management System
TTC	Time-to-Collision
TTI	Time-to-Intersection
V2V	Vehicle-to-Vehicle
V2X	Vehicle-to-Everything
WAAS	Wide Area Augmentation System
WGS	World Geodetic System

5. CONCEPT OF OPERATIONS

5.1 Tutorial [Informative]

A concept of operations describes a proposed system from the users' perspective. Typically, a concept of operations is used on a project to ensure that the system developers understand the users' needs. Within the application standards, the concept of operations documents the purpose of each feature for which an application standard supports an application. The concept of operations also serves as the starting point for users to select those features that may be appropriate for a specific application.

The concept of operations starts with a discussion of the current situation and issues that have led to the need to deploy systems within the scope of a standard and to the development of the standard itself. This discussion permits both potential users and system developers to understand the situation.

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The concept of operations then documents key aspects of the proposed system, including:

a. Reference physical architecture: The reference physical architecture defines the overall context of the proposed system and defines which specific interfaces are addressed.

b. Architectural needs: The architectural needs discuss issues and needs relative to the system architecture.

c. Features: The features identify and describe the various capabilities the users may want that are not currently covered in other standards. These features are derived from the high level user needs identified in the problem statement but are refined and organized into a more manageable structure that forms the basis of the traceability tables contained in 6.3.3.

Architectural needs and features are collectively called user needs. In 6.4, these user needs lead to the definition of the various requirements. Basic systems engineering requires that:

- Each user need traces to one or more functional requirement(s), and
- Each functional requirement derives from at least one user need.

This traceability is shown in the Needs to Requirements Traceability Matrix (NRTM) in 6.3.3.

SAE J2945/1B is intended to be used in class 1 and classes 4 through 13 vehicles as defined by FHWA to ensure that the exchange of BSMs in V2V safety communications provides the desired interoperability and data integrity to support the performance of the envisioned safety applications. Within the NRTM, each user need and requirement is identified as mandatory, optional, or conditional, and users may complete the NRTM to clearly define unique aspects of their implementation. Within SAE J2945/1B, items marked mandatory are those that relate to the most basic functionality required in meeting the identified user needs. For specific implementations, the user identifies those optional or conditional needs appropriate for a specific implementation.

Each requirement is presented in the Requirements Traceability Matrix (RTM) in Appendix A, which defines how the requirement is fulfilled through the standardized dialogs and data entities defined in Sections 7 and 8.

A conformant system may support additional features, as long as it is conformant with the requirements of SAE J2945/1B.

5.2 Current Situation and Problem Statement [Informative]

SAE J2945/1 and SAE J3161/1 address vehicle-safety communications for FHWA classes 2 and 3. With other classes of vehicles on the roads today expecting to have vehicle-to vehicle (V2V) capabilities, it is necessary to address the unique needs of the remaining classes of vehicles. This document specifies the functional and performance requirements for V2V safety communications for FHWA classes 1 and 4 through 13 (non-light-duty), which are not covered in SAE J2945/1 and SAE J3161/1. These specifications provide the requirements necessary to build interoperable systems that support V2V safety applications for non-lightweight vehicles and motorcycles that rely on the exchange of BSMs. The system can transmit and receive the SAE J2735-defined BSM over a wireless communications interface; the communications interface itself is outside the scope of this document.

As shown in Figure 1, motorcycles are class 1 vehicles. Motorcycles, similar to passenger cars, can be represented as a rectangle with certain length and width; however, because of the potential for leaning by the rider, the actual surface space taken up by the vehicle and its rider vary. The information provided from a motorcycle about its physical space it is taking on the road is therefore a function of its position on the road and its varying lean angle, the angle between the center plane of the motorcycle and the vertical plane. To make the necessary data available to other vehicles, SAE J2945/1B sets the requirements for information provided by class 1 vehicles to other connected vehicles/devices.

Class 4, 5, 6, and 7 vehicles, as seen in Figure 1, are all single unit vehicles. Although the size of these vehicles, and their number of axles vary significantly from one class to another, they all can be represented using fixed width and length rectangles. Except for the possibility of towing a trailer or another vehicle, this group—designated as “Rigid Body Group”—do not represent any new requirements beyond those identified in SAE J2945/1.

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Any vehicle used to pull one or a number of trailers, including many of the Rigid Body Group classes and vehicle classes 8 to 13, is hereafter referred to as being a member of the Trailer Group. Typically, the motorized section houses a V2V device, including the positioning devices, while the trailer(s), for the purpose of SAE J2945/1B, are assumed not to be equipped with any V2V technology. For this group of vehicles, there is a need to provide means of providing information about pulling trailers to support safety-related applications, conveying information that is critical to the surrounding vehicles. Information provided by this class of vehicles includes the trailer related data to inform other drivers about:

- Emergency electronic brake lights
- Forward collision warning
- Blind spot warning/lane change warning
- Intersection movement assist
- Control loss warning

For the purposes of defining the user needs, a vehicle that is receiving and acting on BSMs from this group of vehicles is referred to as a host vehicle (HV), the vehicle that is part of this group and transmitting the required data using BSMs is designated a pulling vehicle (PV), and trailer(s) that are not active provider(s) of information are simply referred to as "trailer(s)."

Note that other path constrained vehicles such as articulated buses are supported by this document via the trailer data that is available in the BSM. For example, the vehicle class would still be class 4, but its BSM would include the trailer data to represent the rear section of the articulated bus.

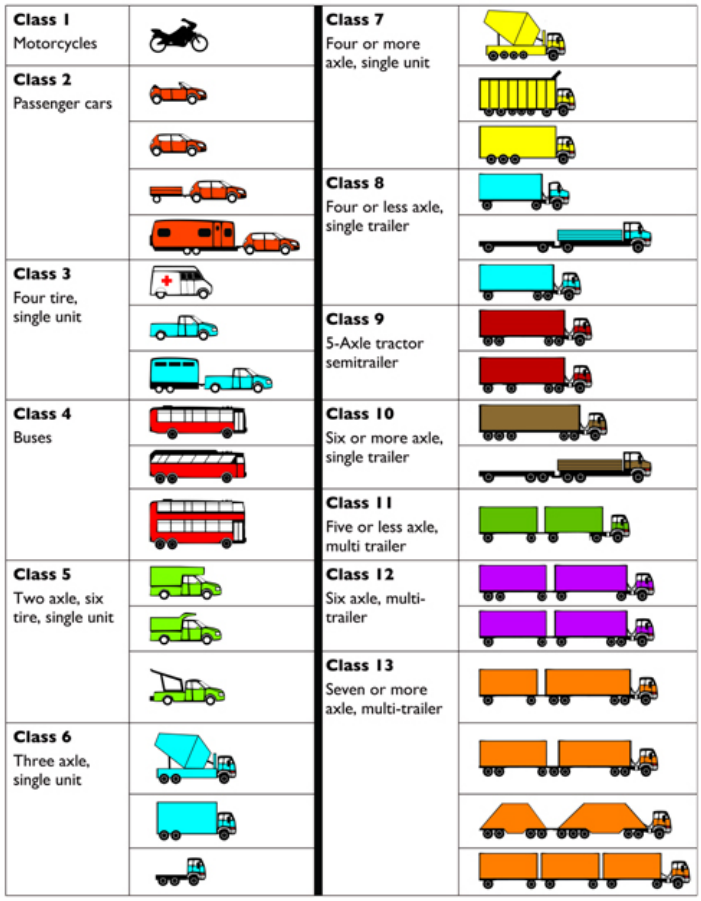

Figure 1 - FHWA vehicle classes (2.2)

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5.3 Reference Architecture

Figure 2 illustrates the components of the V2V system and its interfaces, which is based on the reference physical architecture of SAE J3161/1 and SAE J2945/1.

An infrastructure-based Security Credential Management System (SCMS) is responsible for generating and delivering the security certificates that are used in the message verification process. The SCMS can also revoke certificates that cannot be trusted (e.g., the associated device may have been tampered with or is misbehaving) by placing them on a Certificate Revocation List (CRL) that the SCMS distributes to all systems.

The V2V on-board equipment (OBE), which is the on-board vehicle-to-vehicle (V2V) safety communications system defined in this standard (hereafter referred to as the system), typically consists of multiple subsystem components, which may be discrete or integrated depending on the implementation:

a. V2X radio subsystem: Transmits and receives BSMs. The OBE can include one or more radio subsystems.

b. Positioning subsystem: The subsystem that includes a Global Navigation Satellite System (GNSS) receiver and provides vehicle position, heading, speed, and time information. The system may augment and enhance positioning using additional information and components, which are not shown in Figure 2. Examples of these are speed data from the CAN bus, dead reckoning sensors, optical/camera-based systems, and pivot data.

c. OBE control processor electronic control unit (ECU): Executes software that generates BSMs for transmission according to the requirements in this standard.

d. Antennas: Support radio frequency (RF) links for the V2X radio and GNSS receiver. A second diversity antenna for the V2X radio subsystem is recommended to improve performance. GNSS and radio subsystem antennas may be integrated (dual band).

Systems communicate amongst themselves using the V2X radio subsystem as an interface. The system (OBE) can interface to a Safety Application ECU that detects threats and issues alerts through a driver-vehicle interface (DVI). The DVI can provide visual, audio, and/or haptic alerts. The OBE can also interface with the vehicle controller area network (CAN) bus to obtain vehicle status information. The safety application ECU, CAN bus, and DVI are outside the scope of this document.

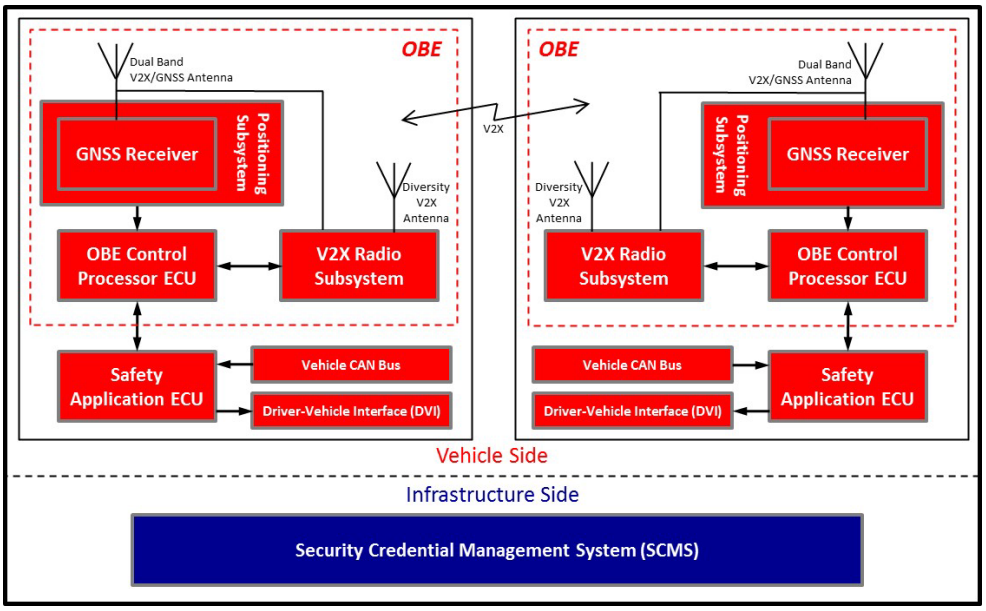

Figure 2 - Reference physical architecture

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5.4 Architectural Needs

5.4.1 V2V Communications Interface

Refer to 5.2.1 of SAE J2945/1 or SAE J3161/1.

5.4.2 Positioning

Refer to 5.2.3 of SAE J2945/1. Note that positioning as described in those documents applies to the pulling vehicle in the case of vehicles with trailers.

5.5 Features

The system needs to provide relevant information so that the following applications, described in SAE J3067, can be supported:

- Forward collision prevention, as described in 2.5.2.1.1 of SAE J3067
- Intersection movement assist, as described in 2.5.2.1.2 of SAE J3067
- Pre-crash actions, as described in 2.5.2.1.3 of SAE J3067
- Control loss warning, as described in 2.5.2.1.4 of SAE J3067
- Lane change warning, as described in 2.5.2.1.5 of SAE J3067
- Do not pass warning, as described in 2.5.2.1.6 of SAE J3067
- Blind spot warning, as described in 2.5.2.1.7 of SAE J3067

5.5.1 Vehicle Class

The system needs to provide the class of vehicle for the vehicle classes 1 and 6 through 13.

5.5.2 Dimensions of Motorcycle in Motion

The area covered by a motorcycle in motion varies based on the height of its rider and the angle the motorcycle's body makes around the longitudinal axis in the direction of motion. The system is to provide information needed to estimate the footprint of a motorcycle in motion. The data includes:

- The roll angle
- Approximate height of a motorcycle in motion

5.5.3 Provide Indication of Presence of Trailer(s)

The system needs to provide an indicator if the vehicle is pulling trailer(s). For an articulated bus, the rigid sections following the engine section can be described as trailers. Note that a motorcycle with a trailer can also be represented using the features in this document.

5.5.4 Indication of Jackknife Condition

A jackknifed vehicle- trailer condition is hazardous to incoming traffic. The system needs to report if a trailer-pulling vehicle encounters a jackknife condition. The conditions that define a jackknife are left to implementers, but initial research suggests that a pivot angle greater than 90 degrees is a good starting point.

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5.5.5 Data Related to Pulling Vehicle

The system needs to provide attributes of the pulling vehicle that are necessary in calculating the position of trailer(s).

5.5.5.1 Provide Indication of Direction of Movement

The estimation of a trailer's position is calculated based on the position of the pulling vehicle (see 5.5.6). The direction of travel is needed in estimating the position of the trailer(s). The system needs to provide an indicator of the movement of a vehicle, forward, neutral, or reverse.

5.5.5.2 Provide Details of Pulling Vehicle Point of Connection

The system needs to provide the physical details that identify its point of connection to a trailer. The information includes:

- Position of the point of connection
- Ability of the point of connection to pivot

5.5.6 Trailer Data for Estimation of a Trailer's Position

The system needs to provide physical trailer attributes of one to eight trailer(s) for a receiving entity to calculate the position of trailer(s) of a class 4 to class 13 vehicle, including:

- Length of each trailer
- Indicator of the ability for the front of a trailer to pivot
- Indicator of the ability for the rear of a trailer to pivot
- Length of the front connection point with respect to the front edge of the trailer
- Length of the rear connection point with respect to the rear edge of the trailer
- Angle a trailer makes with respect to another trailer or vehicle directly in front of it

5.5.7 Other Details for Pulling Vehicle and its Trailer(s)

For the vehicle and each trailer, the system needs to provide:

- Vehicle and trailer height
- Vehicle and trailer width
- Trailer weight
- Vehicle and trailer front and rear bumper height
- Vehicle and trailer total number of axles, and number of front and rear axles

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5.6 Security

All vehicle classes need to implement security as specified in 6.5 and 6.6 of SAE J2945/1. The security solution also needs to include certificate permissions to operate as a class 1 or classes 6 through 13 vehicles, as well as public safety vehicles as described in SAE J3161/1.

5.7 Operational Policies and Constraints

The system must be equipped with a radio communications interface that enables it to send and receive BSMs to and from other vehicles.

5.8 Relationship to the Architecture Reference for Cooperative and Intelligent Transportation (ARC-IT) [Informative]

ARC-IT incorporates all the connected vehicle applications of the Connected Vehicle Reference Implementation Architecture (CVRIA). Physical objects such as vehicles are defined to represent the major physical components of the ITS Architecture. SAE J2945/1B addresses the communications and processing of certain classes of vehicles equipped with platforms aware of their location and their status and capable of communicating with other equipped vehicles.

5.9 Operational Scenarios [Informative]

The operational scenarios do not imply any specific requirements or design. They are provided as examples of user needs to be conveyed to the V2V community.

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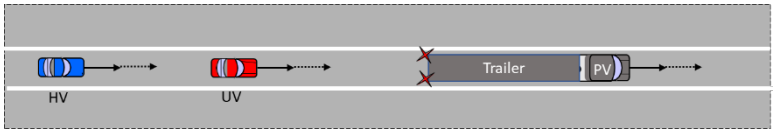
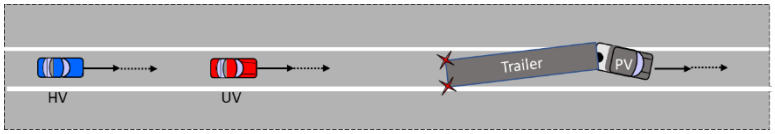
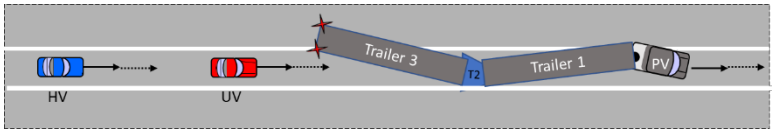
5.9.1 Trailer Group Scenarios

5.9.1.1 Emergency Electronic Brake Lights

Table 1 - Emergency electronic brake lights

Use Case Name	Emergency Electronic Brake Lights.
Category	Safety and Mobility.
Infrastructure Role	Facilitate communication among OBEs.
Short Description	Truck in the same direction of travel of other vehicle(s) abruptly brakes hard. V2V communication is used to warn V2V-equipped vehicles travelling behind the braking truck of the potential danger. The information provided should include the location, speed, and direction of the truck (pulling vehicle [PV]) and each of the trailers it is pulling. The timing of the warning is expected to be set such that the driver of the HV can avoid a rear-end crash with the Unequipped Vehicle (UV) in front (see illustration). The event detected in the PV is to be conveyed from the PV that is ahead and in the same lane, in an adjacent lane, or potentially blocking multiple lanes.
Goal	Avoid accidents and improve flow of traffic.
Constraints	Number of V2V-equipped vehicles. Accuracy of location of cab and trailer as provided by PV.
Geographic Scope	Everywhere.
Actors	V2V-equipped vehicles.
Illustration	 Abruptly slowing PV  Abruptly slowing and swerving PV  Abruptly slowing and swerving PV with multiple trailers
Pre-Conditions	V2V-equipped vehicles running a safety application. Capability of vehicle measuring pivot angle.
Main Flow	The PV transmits information that contains: ▪ A hard-braking event containing the location, speed, and direction of travel of the PV and relevant information about its tailers (accounting for pivot angle as compared to the direction of travel) in the broadcasted BSM during a hard-braking maneuver. Upon receiving such event information, the HV performs the following operations: ▪ Determines which, if any, PVs have reported a hard-braking event. ▪ Determines if it is in a lane effected by the PV (to include trailers) ▪ For each PV that has reported a hard-braking event determines if the distance between the vehicles (including trailers) is less than an implementation-specific threshold value. ▪ Calculates the threat levels among all PVs identified above, determines the principal threat, and sets the appropriate threat status. ▪ Provides a warning to the driver via a driver vehicle interface (DVI).
Alternate Flow(s)	
Post-Conditions	Reduced number of accidents. Improved traffic flow.
Information Requirements	Hard-braking event containing location, speed and direction of the cab, and relevant information about each trailer(s) it is pulling.
Source Documents/References	

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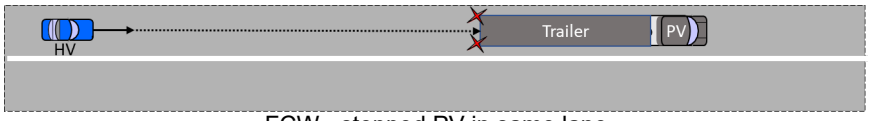
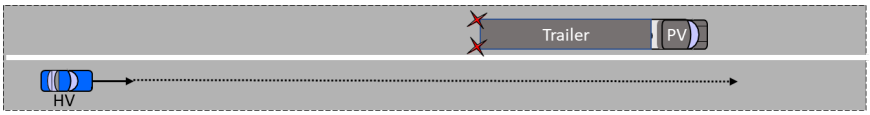
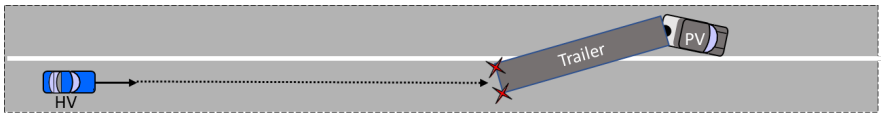
5.9.1.2 Forward Collision Warning

Table 2 - Forward collision warning

Use Case Name	Forward Collision Warning (FCW).
Category	Safety and Mobility.
Infrastructure Role	Facilitate communication among OBEs.
Short Description	The FCW safety application warns the driver of the HV in the case of an impending rear-end crash with an PV and its trailers directly ahead in the same lane and direction of travel. The FCW is intended to help drivers avoid or mitigate rear-end vehicle crashes in the forward path of travel. Cases to be covered include: ▪ Stopped PV+trailers in same lane. ▪ Stopped PV+trailers in adjacent lane. ▪ Slower-moving or decelerating PV+trailers in same lane. ▪ Stopped and obstructing PV+trailers, or jackknifed PV.
Goal	Avoid accidents and improve flow of traffic.
Constraints	Number of V2V-equipped vehicles. Accuracy of provided location and orientation of cab and trailer(s).
Geographic Scope	Everywhere.
Actors	V2V-equipped vehicles.
Illustration	 FCW - stopped PV in same lane  FCW - stopped PV in adjacent lane FCW - slow-moving PV in same lane  FCW - stopped and obstructing PV
Pre-Conditions	V2V-equipped vehicles running a safety application. Capability of vehicle to measure pivot angle.
Main Flow	FCW performs the following operations: ▪ Analyzes received BSMs from each of the PVs and determines which of the PVs are classified as “ahead in-lane(s)” to determine if the HV is at risk of being involved in a rear-end crash with a PV or its connected trailer(s) located in the same lane of travel. ▪ Determines which, if any, PV+trailers classified as “ahead in-lane(s)” are within a range threshold. ▪ Calculates time to collision (TTC) and/or crash avoidance range for each “ahead in-lane” PV+trailers to determine potential forward crash threats. ▪ Identifies the principal threat, if at least one PV or trailer is determined to be a threat. ▪ Provides a warning to the driver via a DVI.
Alternate Flow(s)	
Post-Conditions	Reduced number of accidents. Improved traffic flow.
Information Requirements	FCW containing location, speed and direction of the cab, and each trailer(s) it is pulling.
Source Documents/References	

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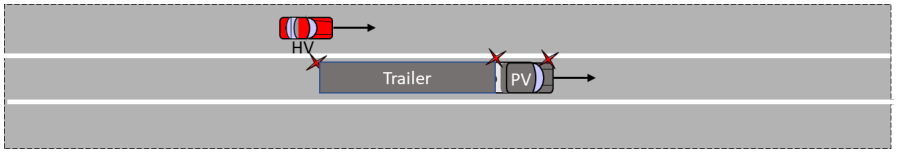
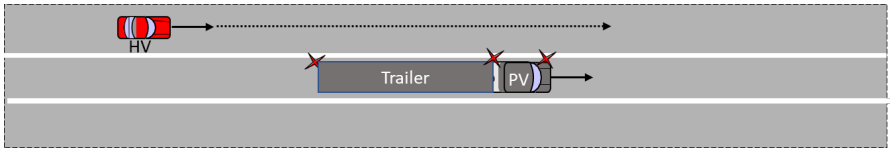
5.9.1.3 Blind Spot Warning/Lane Change Warning

Table 3 - Blind spot warning/lane change warning

Use Case Name	Blind Spot Warning/Lane Change Warning (BSW/LCW).
Category	Safety and Mobility.
Infrastructure Role	Facilitate communication among OBEs.
Short Description	The BSW/LCW safety application warns the driver of the HV during a passing attempt if the blind-spot zone into which the HV intends to move is, or will soon be, occupied by a PV or its trailers traveling in the same direction. Moreover, the application may also provide advisory information that is intended to inform the driver of the HV that a vehicle in an adjacent lane may be attempting a lane change.
Goal	Avoid accidents and improve flow of traffic.
Constraints	Number of V2V-equipped vehicles. Accuracy of provided location and orientation of cab and trailer(s).
Geographic Scope	Everywhere.
Actors	V2V-equipped vehicles.
Illustration	 BSW - PV in blind-spot zone  BSW - approaching PV in adjacent lane
Pre-Conditions	V2V-equipped vehicles running a safety application. Capability of vehicle measuring pivot angle.
Main Flow	BSW/LCW performs the following operations: ▪ Determines if HV has been classified as behind in the adjacent left lane or behind in the adjacent right lane relative to the PV+trailers. ▪ Evaluates the position of HV relative to the PV+trailers to determine if that HV is currently positioned within the PV's blind- spot zone or if that HV is predicted to soon be within the PV's blind-spot zone. ▪ Sets the threat status corresponding to each side (left and right) to advise the HV driver if it is or will be located in the corresponding left or right blind spot of the PV. ▪ Sets the threat status to warn the HV driver of the current or predicted presence of a threat in its lane of travel during a potential lane-change maneuver by an PV. ▪ Provides an advisory or warning to the driver via a DVI.
Alternate Flow(s)	
Post-Conditions	Reduced number of accidents. Improved traffic flow.
Information Requirements	Blind spot warning containing, speed and direction of the cab, and each trailer(s) it is pulling.
Source Documents/References	

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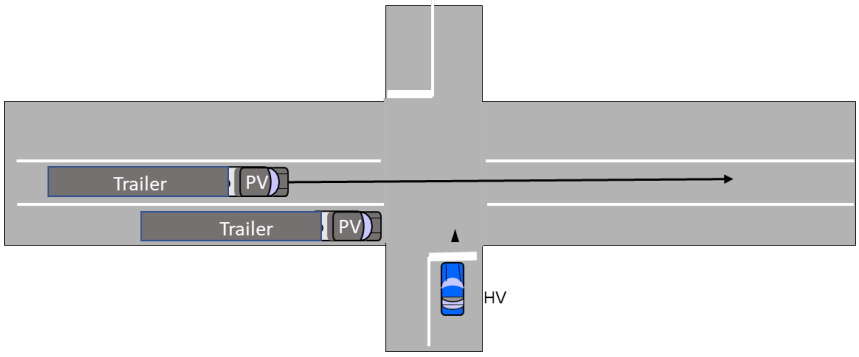
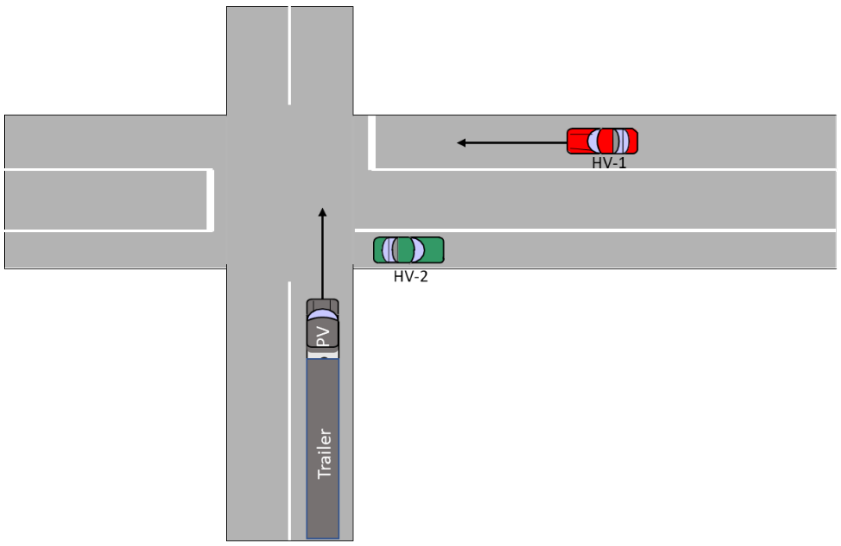
5.9.1.4 Intersection Movement Assist

Table 4 - Intersection movement assist

Use Case Name	Intersection Movement Assist (IMA).
Category	Safety and Mobility.
Infrastructure Role	Facilitate communication among OBEs.
Short Description	The IMA safety application warns the driver of an HV when it is not safe to enter an intersection due to a crash possibility with PV or its trailers. Cases to be covered include: ▪ Stopped HV at intersection. ▪ Both vehicles approaching intersection. ▪ Right turning vehicle blocking lanes. ▪ Left turning vehicle blocking lanes.
Goal	Avoid accidents and improve flow of traffic.
Constraints	Number of V2V-equipped vehicles. Accuracy of provided location and orientation of cab and trailer(s).
Geographic Scope	Everywhere.
Actors	V2V-equipped vehicles.
Illustration	 IMA - stopped HV at the intersection  IMA - both vehicles approaching intersection

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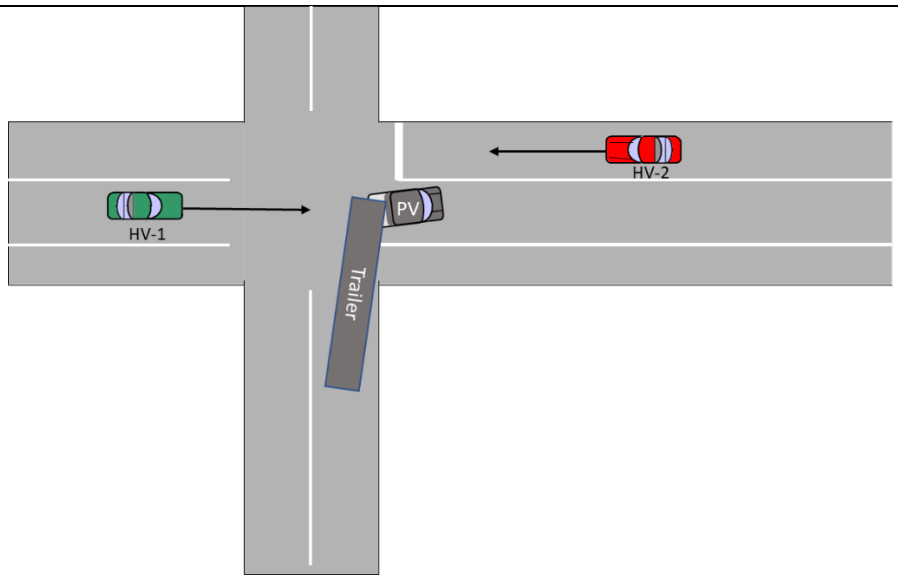
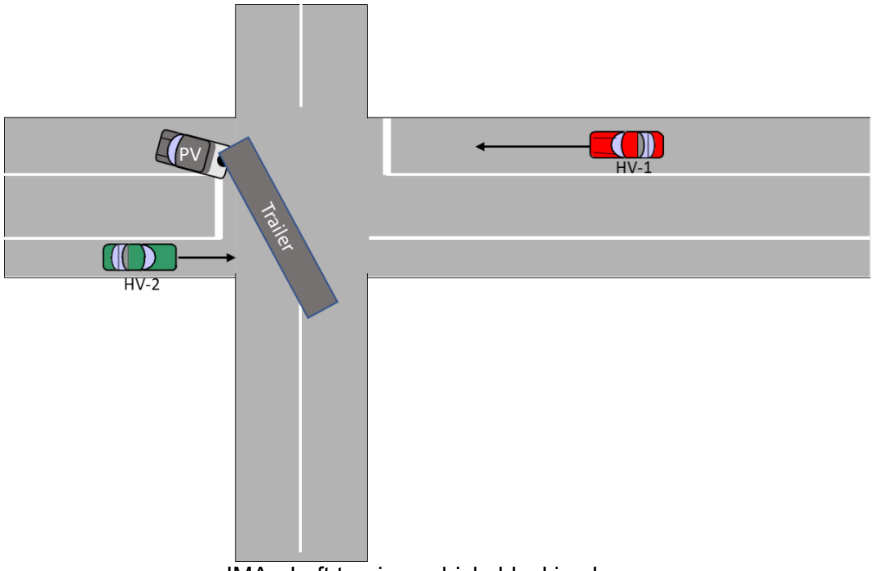
<table>
<tr><td></td><td>


IMA - Right turning vehicle blocking lanes


IMA - Left turning vehicle blocking lanes</td></tr>
<tr><td>Pre-Conditions</td><td>V2V-equipped vehicles running a safety application. Capability of vehicle to measure pivot angle.</td></tr>
<tr><td>Main Flow</td><td>IMA performs the following operations:

▪ Analyzes received BSMs from PVs approaching the intersection
▪ Calculates time-to-intersection (TTI) and distance-to-intersection (DTI)
▪ Identifies the principal threat, if at least one PV is determined to be a threat
▪ For each PV “intersecting left” or “intersecting right,” determine if the accompanying PV+trailers may cause a risk to a HV crossing the intersection.
▪ Provides a warning to the driver via a DVI.</td></tr>
<tr><td>Alternate Flow(s)</td><td></td></tr>
<tr><td>Post-Conditions</td><td>Reduced number of accidents. Improved traffic flow.</td></tr>
<tr><td>Information Requirements</td><td>Location, speed and direction of the cab, and each trailer(s) it is pulling.</td></tr>
</table>

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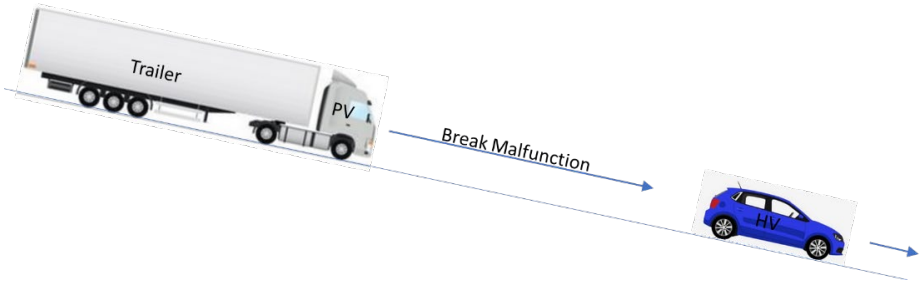
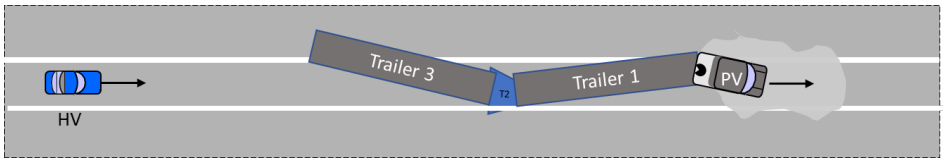
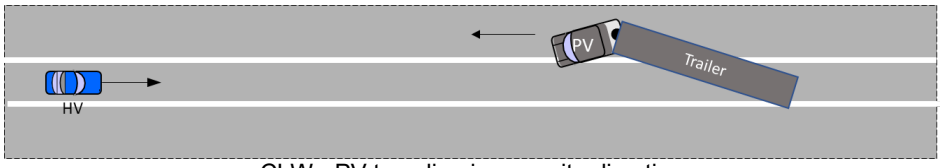
5.9.1.5 Control Loss Warning

Table 5 - Control loss warning

Use Case Name	Control Loss Warning (CLW).
Category	Safety and Mobility.
Infrastructure Role	Facilitate communication among OBEs.
Short Description	The CLW safety application warns the driver of an HV in the case of an emergency control loss event (defined as failure of brakes, or activation of the antilock brake system, traction control system, or stability control system) by an PV traveling in the same or opposite direction. The PV broadcasts control loss event information within the BSM. Upon receiving such event information, the HV determines the relevance of the event and provides a warning to the driver of the HV. Cases include: ▪ PV follows HV in the same direction. ▪ HV follows the PV in the same direction. ▪ PV Traveling in opposite direction. Each truck tractor manufactured on or after March 1, 1997 (except truck tractors engaged in driveaway-towaway operations) shall be equipped with an antilock brake system that meets the requirements of FMVSS No. 121 (49 CFR 571.121, S5.1.6.1(b)).
Goal	Avoid accidents and improve flow of traffic.
Constraints	Number of V2V-equipped vehicles. Accuracy of provided location and orientation of cab and trailer(s).
Geographic Scope	Everywhere.
Actors	V2V-equipped vehicles.
Illustration	 CLW - PV follows HV in the same direction  CLW - HV follows the PV in the same direction  CLW - PV traveling in opposite direction
Pre-Conditions	V2V-equipped vehicles running a safety application. Capability of vehicle to measure pivot angle.

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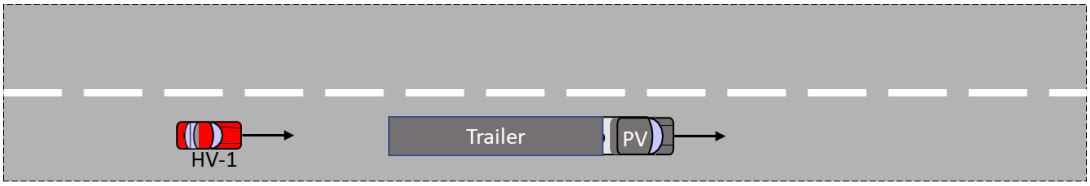
Main Flow	PV detects a defined loss of control event including higher than normal rate of acceleration, fault of the Brake system, activation of the antilock brake system, traction control system, or stability control system. PV broadcasts a control loss event in the BSM along with relevant information for its trailers. Upon receiving such event information, the HV performs the following operations: ▪ Determines which, if any, PVs have reported a control loss event. ▪ For each PV that has reported an event determines if the range between HV and PV and its trailers is less than a threshold value. ▪ Calculates the CLW threat levels among all PVs identified above, determines the principal threat, and sets the appropriate threat status. ▪ Provides a warning to the HV driver via a DVI.
Alternate Flow(s)	
Post-Conditions	Reduced number of accidents. Improved traffic flow.
Information Requirements	Location, speed and direction of the cab, and each trailer(s) it is pulling.

5.9.1.6 Do Not Pass Warning

Table 6 - Do not pass warning

Use Case Name	Do Not Pass Warning (DNPW).
Category	Safety and Mobility.
Infrastructure Role	Facilitate communication among OBEs.
Short Description	The DNPW safety application warns the drivers of HVs following an PV+trailers of potential danger in passing the PV. The PV broadcasts DNPW information within the BSM. Upon receiving such event information, the HV determines the relevance of the event and provides a warning to the driver of the HV.
Goal	Avoid accidents and improve flow of traffic.
Constraints	Number of V2V-equipped vehicles. Accuracy of provided location and orientation of cab and trailer(s).
Geographic Scope	Everywhere.
Actors	V2V-equipped vehicles.
Illustration	 DNPW - V2V enabled vehicles
Pre-Conditions	V2V-equipped vehicles running a safety application. Capability of vehicle to measure pivot angle.
Main Flow	PV detects a change in its movement that may prohibit an HV from passing it. PV broadcasts a do not pass warning event along with the relevant information of its trailers in the BSM. Upon receiving such event information, the HV performs the following operations: ▪ Determines which, if any, PVs have reported a do not pass warning. ▪ For each PV+trailers that has reported an event determines if the range is less than a threshold value. ▪ Calculates the passing threat levels among all PVs identified above, determines the principal threat, and sets the appropriate threat status. ▪ Provides a warning to the driver via a DVI.
Alternate Flow(s)	
Post-Conditions	Reduced number of accidents. Improved traffic flow.
Information Requirements	Location, speed and direction of the cab, and each trailer(s) it is pulling. Location, speed, and direction of incoming vehicles.

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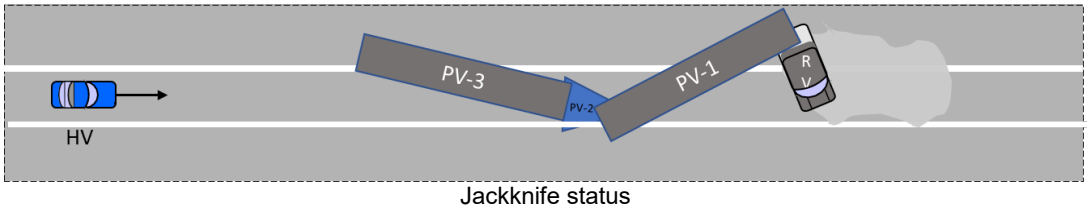
5.9.1.7 Jackknife Status

Table 7 - Jackknife status

Use Case Name	Jackknife Status.
Category	Safety and Mobility.
Infrastructure Role	Facilitate communication among OBEs.
Short Description	The safety application warns the drivers of HVs around an PV that has jackknifed.
Goal	Avoid accidents and improve flow of traffic.
Constraints	Number of V2V-equipped vehicles. Accuracy of provided location and orientation of cab and trailer(s).
Geographic Scope	Everywhere.
Actors	V2V-equipped vehicles.
Illustration	 Jackknife status
Pre-Conditions	V2V-equipped vehicles running a safety application. Capability of vehicle to measure pivot angle.
Main Flow	PV detects a jackknife condition and broadcasts this status along with the relevant information of its trailers to surrounding HVs. Upon receiving such event information, the HV performs the following operations: ▪ Determines which PV has reported the status. ▪ For each PV+trailers that has reported an event determines if the range is less than a threshold value. ▪ Calculates the passing threat levels among, determines the principal threat, and sets the appropriate threat status. ▪ Provides a warning to the driver via a DVI.
Alternate Flow(s)	
Post-Conditions	Reduced number of accidents. Improved traffic flow.
Information Requirements	Location and direction of the cab, and each trailer(s) it is pulling.

6. FUNCTIONAL REQUIREMENTS

6.1 Tutorial [Informative]

Refer to SAE J2945.

6.2 Scope of the Interface [Informative]

Refer to SAE J2945.

6.3 Needs to Requirements Traceability Matrix (NRTM)

6.3.1 Notation [Informative]

Information about NRTM notation may be found in NTCIP 8002 Annex B1 section 3.3 and SAE J2945.

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6.3.1.1 Conformance Symbols

The following symbols are used to indicate status in the conformance column of the NRTM table:

Table 8 - Conformance symbols

Symbol	Value
M	Mandatory: The system is required to implement this functionality.
M.#	Every item of the group labeled by the same numeral # is mandatory, but only one is active at a time.
O	Optional: The system does not need to implement the feature to be standards compliant.
O.# (range)	Part of an option group: At least the number of options indicated by the "(range)" is required from the options labeled with the same numeral #. (1)
C	Conditional: The system must implement functionality corresponding to this requirement if another requirement is implemented.
NA	Not-applicable (i.e., logically impossible in the scope of the standard).
X	Excluded or prohibited.

(1) The O.# (range) notation is used to show a set of selectable options (e.g., O.2 (1..*)) would indicate that one or more of the option group two options must be implemented. Two character combinations are used for dynamic requirements. In this case, the first character refers to the static (implementation) status, and the second refers to the dynamic (use); thus, .MO. means mandatory to be implemented, optional to be used.

6.3.1.2 Conditional Status Notation

The following predicate notation may be used:

Table 9 - Predicate notations

<predicate>:	This notation introduces a single item that is conditional on the <predicate>.
<predicate>::	This notation introduces a table or a group of tables, all of which are conditional on the <predicate>.
(predicate)	This notation introduces the first occurrence of the predicate either in the NRTM table or that specific user need. The feature associated with this notation is the base feature for all options that have this predicate in their conformance column.

The <predicate>: notation means that the status following it applies only when the NRTM states that the feature or features identified by the predicate are supported. In the simplest case, <predicate> is the identifying tag of a single NRTM item. When the group predicate is true, then the associated section shall be completed. The symbol <predicate> also may be a Boolean expression composed of several indices. .AND., .OR., and .NOT. shall be used to indicate the Boolean logical operations.

The predicates used in this document are as follows:

- Motorcycle: Vehicle is identified as a FHWA class 1 vehicle (motorcycles)
- Trailer: Trailer Present indicator is set to show trailer is present

6.3.1.3 Support Column Symbols

The support column can be used by a procurement specification to identify the required features for the given procurement or by an implementer to identify which features have been implemented. In either case, the user circles the appropriate answer (yes, no, or NA) in the support column:

Table 10 - Support column symbols

Yes	To be supported (or supported) by the implementation.
No	Does not have to be supported (or not supported) by the implementation.
NA	Not applicable.

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6.3.2 Instructions for Completing the NRTM [Informative]

To claim conformance with this standard, an implementation must satisfy the mandatory and selected optional requirements as identified in the NRTM.

The columns in the NRTM table are the user need ID and user need title columns, FR ID (functional requirement ID) and FR title (functional requirement title) columns, the conformance column, the support column, and the additional specifications column.

6.3.2.1 User Need ID and User Need Title Columns

The NRTM is based upon the user needs defined in each section. The section number and user need name are indicated within these columns.

6.3.2.2 FR ID and FR Title Columns

The requirements are defined within 6.4 to 6.6. The NRTM references the traces from user needs to these requirements. The section number and requirements name are indicated within these columns.

6.3.2.3 Conformance Column

The notations and symbols used to indicate status and conditional status in the NRTM, for this column are defined in 6.3.1.1 and 6.3.1.2. Not all of these notations and symbols may be used in this document.

6.3.2.4 Support Column

In the “support” column, each response shall be selected either from the indicated set of responses (for example: yes, no, NA), or it shall reference additional items that are to be attached.

If a conditional requirement is inapplicable, use the not applicable (NA) choice.

6.3.2.5 Additional Specifications Column

The additional specifications column may (and should) be used by a procurement specification to provide additional notes and requirements for the product to be procured or may be used by an implementer to provide any additional details about the implementation. In some cases, default text already exists in this field, which the user should complete in order to fully specify the implementation. However, additional text can be added to this field as needed to fully specify a feature. Performance requirements should also be specified in this column.

6.3.3 Needs to Requirements Traceability Matrix (NRTM) Table

To claim conformance with this standard, an implementation shall satisfy the mandatory and selected optional requirements as identified in the NRTM.

Table 11 - NRTM table

User Need ID	User Need	FR ID	Functional Requirement	Conformance	Support	Additional Specifications
5.4	Architectural Needs					
		6.4	Architectural Requirements	M	Yes	SAE J2945/1, SAE J3161/1
5.4.1	V2V Communications Interface					
		6.4.1	Communications Requirements	M	Yes	SAE J2945/1, SAE J3161/1
5.4.2	Positioning					
		6.4.2	Positioning Requirements	M	Yes	SAE J2945/1
5.5	Features					
		6.5	Features			
5.5.1	Vehicle Class					
		6.5.1	Non-Light-Duty Vehicle Class	M	Yes	SAE J2945/1

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User Need ID	User Need	FR ID	Functional Requirement	Conformance	Support	Additional Specifications
5.5.2	Dimensions of Motorcycle in Motion					
		6.5.2	Motorcycle Requirements	Motorcycle: O	Yes/No	
		6.5.2.1	Motorcycle Roll Angle	Motorcycle: O	Yes/No	
		6.5.2.2	Motorcycle Reported Height	Motorcycle: O	Yes/No	
5.5.3	Provide Indication of presence of Trailer(s)					
		6.5.3.1.2	Trailer Present Indication	O	Yes/No	
5.5.4	Indication of Jackknife Condition					
		6.5.3.1.3	Jackknife Flag	O	Yes/No	
5.5.5	Data related to pulling vehicle					
		6.5.3	Vehicles with Trailers Requirements	O	Yes/No	
		6.5.3.1	Pulling Vehicle Attributes	O	Yes/No	
5.5.5.1	Provide Indication of Direction of Movement					
		6.5.3.1.1	Transmission State	M	Yes	SAE J2945/1
5.5.5.2	Provide Details of pulling vehicle point of connection					
		6.5.3.1.4	Pivot Ability	O	Yes/No	
		6.5.3.1.5	Pivot Offset	O	Yes/No	
5.5.6	Trailer Data for estimation of a Trailer's Position					
		6.5.3.2	Information about Trailers	Trailer: O	Yes/No	
		6.5.3.2.1	Trailer Length	M	Yes	
		6.5.3.2.2	Trailer Front Pivot Ability	M	Yes	
		6.5.3.2.3	Trailer Rear Pivot Ability	O	Yes/No	
		6.5.3.2.4	Trailer Front Pivot Position Relative to Trailer's Front Bumper	M	Yes	
		6.5.3.2.5	Trailer Rear Pivot Position Relative to Trailer's Rear Bumper	O	Yes/No	
		6.5.3.2.6	Trailer Pivot Angle	M	Yes	
5.5.7	Other Details for Pulling vehicle and its trailer(s)					
		6.5.3.1.9	Vehicles with Trailers Reported Height	O	Yes/No	
		6.5.3.1.10	Vehicles with Trailers Reported Width	M	Yes	SAE J2945/1
		6.5.3.1.6	Pulling Vehicle total number of Axles	O	Yes/No	
		6.5.3.1.7	Pulling Vehicle number of Front Axles	O	Yes/No	
		6.5.3.1.8	Pulling Vehicle number of Rear Axles	O	Yes/No	
		6.5.3.2.7	Trailer Height	Trailer: O	Yes/No	
		6.5.3.2.8	Trailer Width	Trailer: M	Yes	
		6.5.3.2.9	Trailer Weight	Trailer: O	Yes/No	
		6.5.3.2.10	Trailer Front Bumper Height	Trailer: O	Yes/No	
		6.5.3.2.11	Trailer Rear Bumper Height	Trailer: O	Yes/No	
		6.5.3.2.12	Trailer total number of Axles	Trailer: O	Yes/No	
		6.5.3.2.13	Trailer number of Front Axles	Trailer: O	Yes/No	
		6.5.3.2.14	Trailer number of Rear Axles	Trailer: O	Yes/No	
5.6	Security					
		6.6	Security	M	Yes	Appendix B, SAE J3161/1, SAE J2945/1

6.4 Architectural Requirements

This section provides the details of the information to be provided in BSMs transmitted by non-light-duty vehicles. Unless specified in this document, all non-light-duty vehicles shall comply with the BSM contents and performance requirements as specified in SAE J2945/1 or 6.1.6, 6.3.1, and 6.3.5 of SAE J3161/1. In the case of vehicles with trailers, the trailer information is added to the BSM as described in this document (SAE J2945/1B). Note that in the case of public safety vehicles, these vehicles are permitted to use additional SAE J2735 features as described in 6.1.6 of SAE J3161/1. However, SAE J3161 is limited to support for public safety vehicle classes 2 through 5. For implementations conformant with this document, any class 1 or classes 6 through 13 public safety vehicles with the proper service specific permissions are conformant with the requirements in 6.1.6 of SAE J3161/1. Class 2 through 5 public safety vehicles are already addressed by SAE J3161/1 and not considered in this document.

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6.4.1 Communications Requirements

Communications requirements are outside the scope of this document. Refer to SAE J2945/1 and SAE J3161/1 for communications interface requirements, including congestion control as noted in 6.3.8 of those documents.

6.4.2 Positioning Requirements

The system shall comply with the requirements of 6.2 of SAE J2945/1. The vehicle position (position reference) reported in a BSM shall be a point (latitude, longitude, and elevation) projected onto the surface of the roadway (road plane) with reference to the WGS-84 coordinate system and its reference ellipsoid. This point is the center of the rectangle on the road plane, oriented about the vehicle that encompasses the farthest forward, rearward, and side-to-side points on the vehicle, including original equipment such as outside side view mirrors and without any consideration to presence of any trailer(s) (see Figure 3). Trailers are assumed not to be equipped with V2V devices. The position of each trailer, if available, is derived from the position of the V2V-equipped pulling vehicle through connection points between the pulling vehicle and the trailer, or between trailers in the case of a multiple trailers.

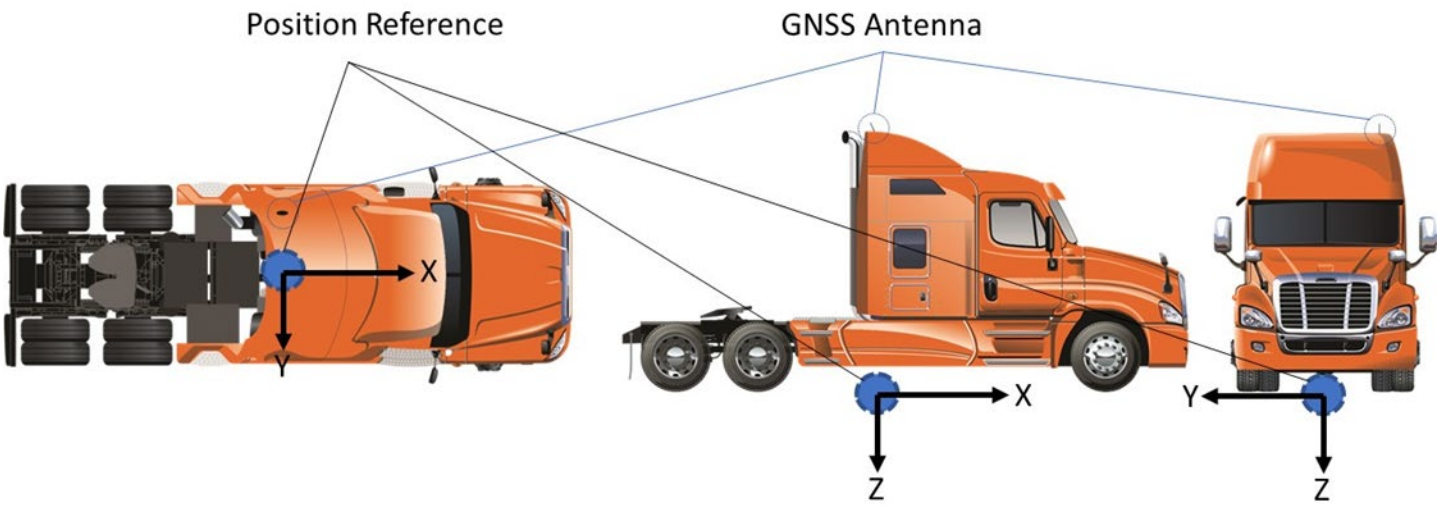

Figure 3 - Vehicle position reference

In case of vehicles with trailers, the position reference is that of the vehicle itself regardless of presence of any trailer(s).

6.5 Features

All vehicles, regardless of FHWA vehicle class, shall comply with 6.1.6, 6.3.1, 6.3.5, 6.3.6, and 6.3.7 of SAE J2945/1. Public safety vehicles shall also comply with the supplemental requirements of 6.1.6 of SAE J3161/1. Motorcycles and vehicles with trailers have additional requirements specified in this section.

Note that the some of the vehicle classes identified in this document (e.g., trucks with trailers) may have dynamic characteristics. This includes changes in width due to adjustable mirrors or adjustable features on trailers. Weight may also change or shift during a trip. These adjustable parameters are supported by the BSM, but the means by which adjustments are made are outside the scope of this document.

6.5.1 Non-Light-Duty Vehicle Class

All non-light-duty vehicles shall include FHWA vehicle class (1 or 6 through 13) in the BSM.

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6.5.2 Motorcycle Requirements

6.5.2.1 Motorcycle Roll Angle

The system shall include the roll angle of the motorcycle in its BSM.

Roll angle, shown in Figure 4, is defined as the angle a motorcycle's body makes around the longitudinal axis in the direction of motion without considering the position of the rider. The roll angle is measured in degrees from -90 to +90, with the angle to the right of the longitudinal axis defined as positive.

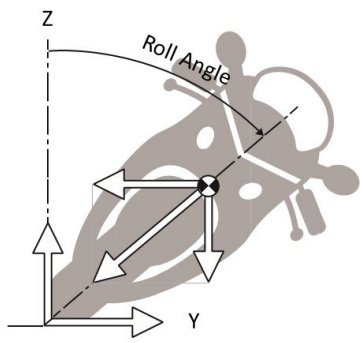

Figure 4 - Motorcycle roll angle

6.5.2.2 Motorcycle Reported Height

The system shall include a height equal to the height of the motorcycle plus 1.0 m. The additional length is an estimate for the height of the rider.

6.5.3 Vehicles with Trailers Requirements

This section includes requirements for vehicles with trailers.

6.5.3.1 Pulling Vehicle Attributes

The pulling vehicle attributes shall at a minimum include the indication of presence of trailer(s).

6.5.3.1.1 Transmission State

The system shall include an indicator of the state of the vehicle's transmission ("forward," "reverse," "neutral") in accordance to 6.3.6.9 of SAE J2945/1.

6.5.3.1.2 Trailer Present Indication

The system shall include an indication of presence of trailer(s) ("present," "not present").

6.5.3.1.3 Jackknife Flag

The system shall include an indicator signifying a jackknife condition.

6.5.3.1.4 Pivot Ability

The system shall include an indicator of pivot ability of the connection point of the pulling vehicle ("true," "false").

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6.5.3.1.5 Pivot Offset

The system shall include the position of the connection point where a trailer connects to the pulling vehicle. The position is measured relative to the back of the pulling vehicle, along the centerline of its outline. Shown in Figure 5, R_A, representing the position of the connection point, is the distance between the back of the pulling vehicle to the center of the connection point (hitch). The value is represented as a negative number if the connection point is on the body of the vehicle, such as the hitch in Figure 5, a positive number if the connection point extends beyond the length of the pulling vehicle, effectively elongating the vehicle, or zero (0) if the point of connection is right on the back bumper.

Figure 5 - Pulling vehicle's connection point

6.5.3.1.6 Pulling Vehicle Total Number of Axles

The system shall include the total count of the axles of the pulling vehicle.

6.5.3.1.7 Pulling Vehicle Number of Front Axles

The system shall include the count of front axles of the pulling vehicle.

6.5.3.1.8 Pulling Vehicle Number of Rear Axles

The system shall include the count of rear axles of the pulling vehicle.

6.5.3.1.9 Vehicles with Trailers Reported Height

The system shall include the height of the highest part of the pulling vehicle as its reported height, measured from the ground to the highest surface, excluding any antenna(s), as shown in Figure 6. Under certain road conditions (e.g., ice), actual vehicle height may vary; however, reported vehicle height does not change based on road conditions.

Figure 6 - Vehicle with trailers: reported height

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6.5.3.1.10 Vehicles with Trailers Reported Width

The system shall include the width of the widest section of the pulling vehicle. Note that some pulling vehicles and trailers have features that result in varying vehicle/trailer width (e.g., adjustable mirrors). An HMI may be required for a vehicle operator to enter system parameters in this case. Alternatively, the information could be written to the vehicle system by roadside measurement systems. The means by which this is done is outside the scope of this document.

6.5.3.2 Information about Trailers

The system shall include descriptive information for one to eight trailers attached to a vehicle.

A trailer is defined as a component capable of being moved in the direction of movement of the pulling vehicle. A trailer may have a maximum of two points of pivot. Figure 7 shows trailers with a single point of pivot in the front.

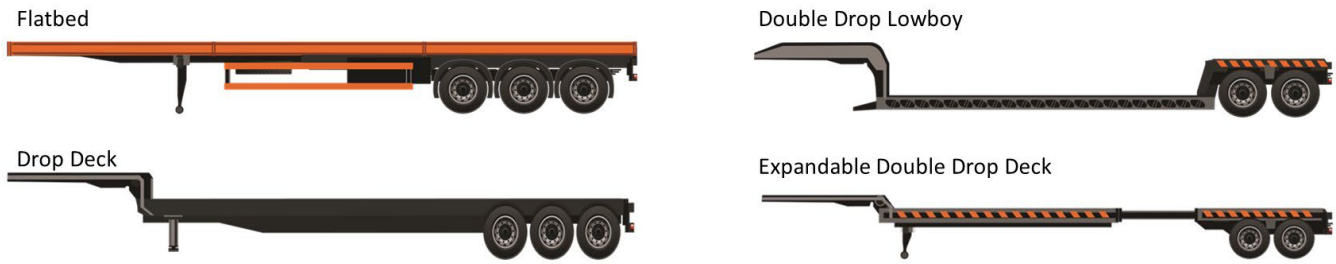

Figure 7 - Trailer with single point of pivot

The trailer in Figure 8 has a point of pivot in the front plus an additional dolly. The dolly has two points of pivot, one in front of the dolly and one in the back, thus leading to the need for a two-trailer representation. The trailer in Figure 8 is therefore to be represented with descriptive information for two trailers.

Figure 8 - Two-trailer configuration

As shown in Figure 9, due to the presence of multiple points of pivot, the expandable trailer is represented as being three trailers. Similarly, as shown in Figure 10, the presence of the connecting dolly (between the two tankers) with two points of pivot, leads to a three-trailer representation for the tanker truck.

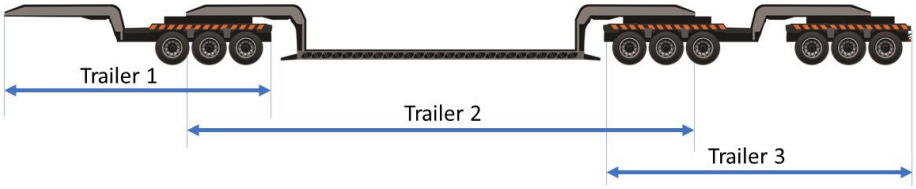

Figure 9 - Three-trailer configuration

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Figure 10 - Three-trailer configuration (two tankers and a dolly)

The attributes are used by a receiving entity to estimate the position of each trailer. The relative attributes, always measured in reference to the vehicle or trailer immediately in front of the trailer of interest, describe the geometric relationship between a pulling vehicle and the first trailer, dolly, or any other object to which it is connected to, as well as the geometric relationship between each trailer, if any, and the trailer immediately ahead of it.

The attributes include measurement of the location of each point of connection that exist in the connected chain. A point of connection can be fixed (non-pivoting) or can rotate in the horizontal plane at the connection point. The connection point is presumed to be along the centerline of the object in question. While information about rotation in the horizontal plane is key in calculating the position of each trailer, rotation in the vertical plane (pitch and roll) is not critical.

The distance offset of a point of connection, the pivot offset, is with respect to the length and tangential to the width of the object in question. The current angle between two objects (expressed with respect to the pulling vehicle or another trailer, whichever is directly in front) is a dynamic value when the vehicle is in motion. The measurement provided for a trailer is with respect to the object in front of it. Only the pulling vehicle position and heading angle are absolute.

6.5.3.2.1 Trailer Length

Trailer information shall include the length of the trailer measured from the edge of the front bumper to the edge of the rear bumper.

Each trailer in the chain will have its length included in the information provided by the vehicle. As depicted in Figure 11, Trailer 2 is connected to Trailer 1 using a fixed connection with the pivot point at the rear end of Trailer 1. In this case, the trailers of this vehicle are described using information for two trailers, where L_B is the length of Trailer 1 and L_C is the length of Trailer 2. The length of the connecting hitch, connecting Trailer 1 to Trailer 2, is accounted for as an attaching component (see 6.5.3.2.4). The pulling vehicle's length, L_A, is provided in the BSM Coredata (commonly referred to as Part I), while the lengths of the two trailers will be provided as part of BSM Part II.

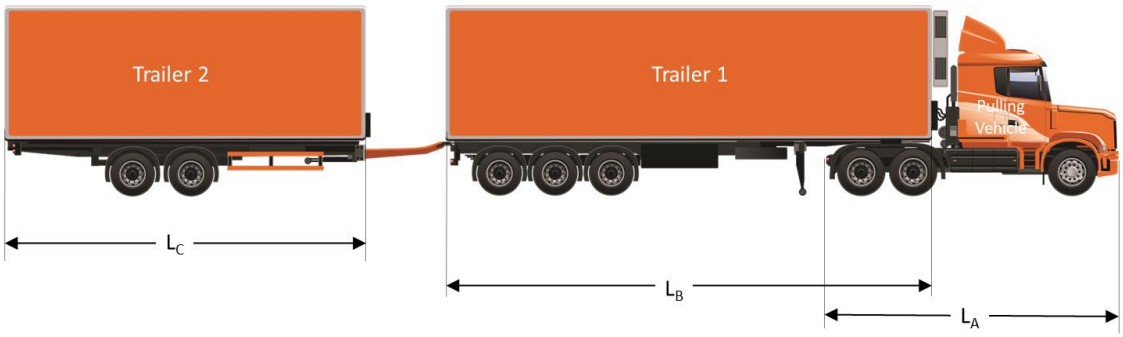

Figure 11 - Lengths of trailers: two-trailer configuration

In Figure 12, Trailer 2 is connected to Trailer 1 using a dolly (Trailer 2) with two points of pivot. The presence of two points of pivot leads to including the dolly information leading to this three-trailer configuration with each trailer having its own length, L_B for Trailer 1, L_C for Trailer 2 (the dolly), and L_D for Trailer 3.

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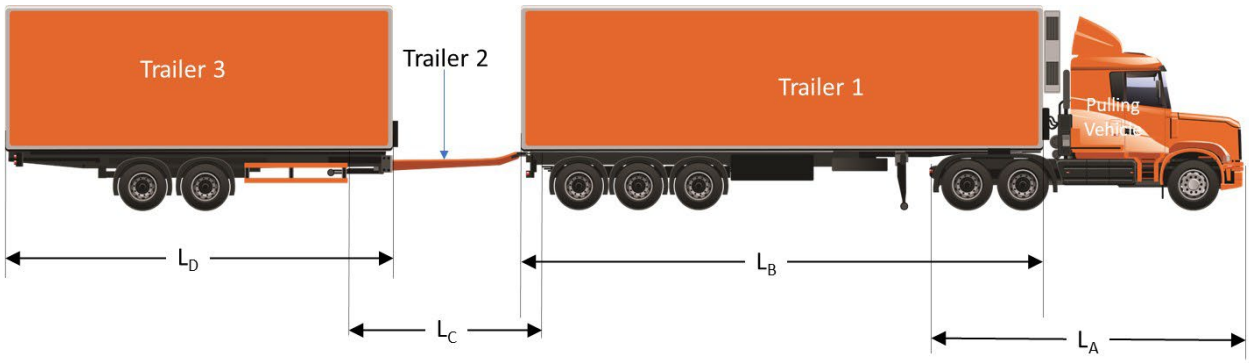

Figure 12 - Lengths of trailers: three-trailer configuration

6.5.3.2.2 Trailer Front Pivot Ability

For each trailer, the system shall include an indication of pivot ability of the front of the trailer with respect to the vehicle or another trailer that is immediately in front (“true,” “false”).

6.5.3.2.3 Trailer Rear Pivot Ability

For each trailer, the system shall include an indication of pivot ability of the rear of the trailer with respect to the trailer that is immediately behind it (“true,” “false”).

6.5.3.2.4 Trailer Front Pivot Position Relative to Trailer’s Front Bumper

The system shall include the position of its front attachment point along the centerline of the trailer with a positive value indicating extension beyond the outline of the trailer and a negative value indicating that the pivot position is positioned on the outline trailer.

As depicted in Figure 13, Trailer 1’s front pivot point is at a distance F_B and positioned on the outline of Trailer 1, therefore its value is represented by a negative number. Trailer 2’s single pivot point, located in front of the trailer, extends beyond Trailer 2’s boundary by a distance F_c and effectively adds to the length of the trailer thus represented by a positive value.

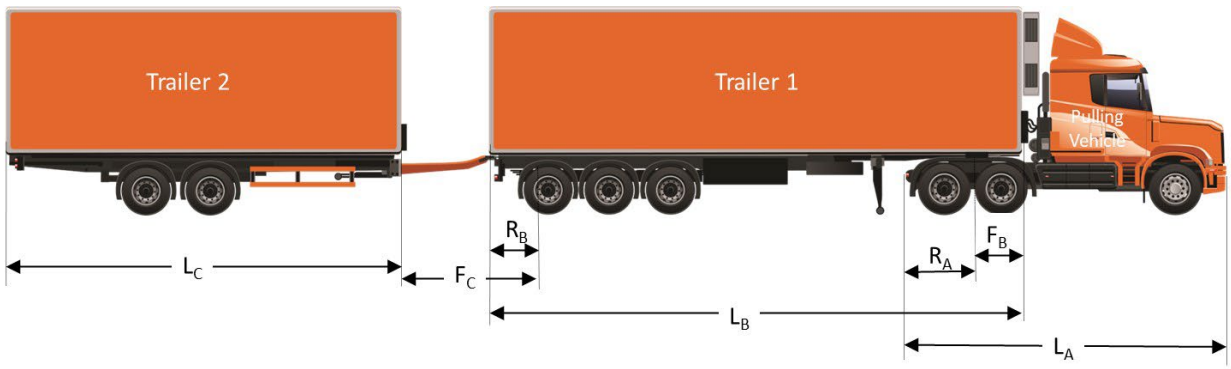

Figure 13 - Pivot point offsets

6.5.3.2.5 Trailer Rear Pivot Position Relative to Trailer’s Rear Bumper

The system shall include the position of its rear attachment point along the centerline of the trailer with a positive value indicating extension beyond the outline of the trailer, and a negative value indicating that the pivot position is positioned within the outline of the trailer.

Referring to Figure 13, Trailer 1’s rear pivot point is at a distance R_B and positioned on the outline of Trailer 1; therefore, its value is represented by a negative numbers. Trailer 2 does not have a rear pivot point.

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6.5.3.2.6 Trailer Pivot Angle

The system shall include the angle of a trailer with respect to the trailer or pulling vehicle directly in front of it. The pivot angle is the difference between the heading of the subject trailer and the trailer or pulling vehicle in front of it.

Angle of interest is the angle between the center-line of the trailer of interest and the trailer directly ahead or a vehicle which is pulling it. The angle is a left-hand angular rotation with respect to the direction of the movement of the vehicle or trailer directly ahead. As an example, the angle, α_A, shown in Figure 14, is represented by a negative value. A value of 0 indicates movement in the same direction as the trailer or vehicle directly in front of the trailer of interest.

In a multi-trailer configuration, as shown in Figure 15, α_B is the angle the first trailer makes with respect to the pulling vehicle, α_C is the angle the Trailer 2 (dolly) makes with respect to the first trailer, and α_D is the angle the Trailer 3 makes with respect to Trailer 2. Following the convention described earlier, α_B will have a negative value while α_C and α_D will have positive values.

The receiving entity may calculate the pivot angle(s) by using the heading of each of the connected components.

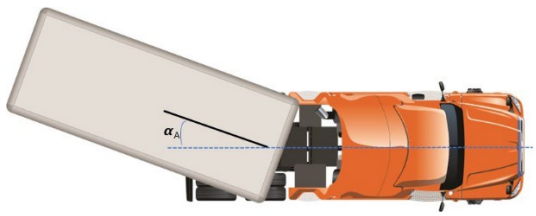

Figure 14 - Pivot angle

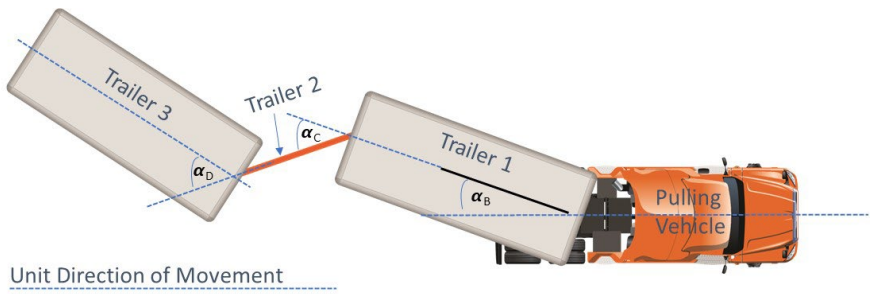

Figure 15 - Pivot angles: multi-trailer configuration

As shown in Figure 16, articulated buses are treated as vehicles with trailers, with the front bus considered as the pulling vehicle and the rear section as a trailer.

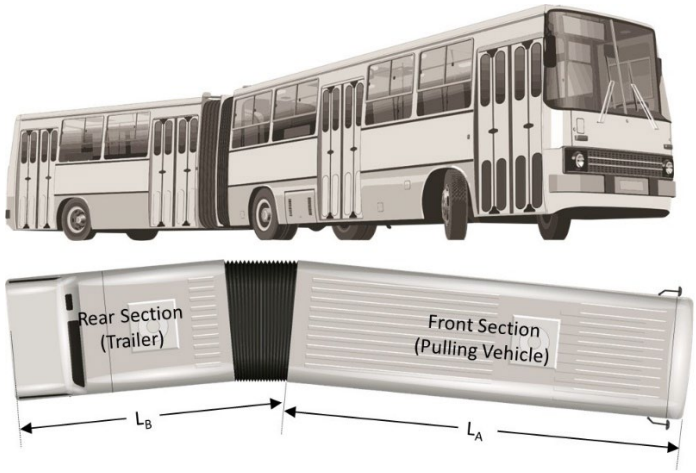

Figure 16 - Articulated bus

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6.5.3.2.7 Trailer Height

The system shall include the height of the highest point of the trailer.

6.5.3.2.8 Trailer Width

The system shall include the width of the widest section of the trailer.

6.5.3.2.9 Trailer Weight

The system shall include the current weight of the trailer.

6.5.3.2.10 Trailer Front Bumper Height

The system shall include the height of the front bumper of the trailer.

In cases of vehicles with complex bumper shapes, the center of the mass of the bumper (where the bumper can best absorb an impact) should be used.

6.5.3.2.11 Trailer Rear Bumper Height

The system shall include the height of the rear bumper of the trailer.

In cases of vehicles with complex bumper shapes, the center of the mass of the bumper (where the bumper can best absorb an impact) should be used.

6.5.3.2.12 Trailer Total Number of Axles

The system shall include the total count of the axles of the trailer.

6.5.3.2.13 Trailer Number of Front Axles

The system shall include the count of front axles of the trailer.

6.5.3.2.14 Trailer Number of Rear Axles

The system shall include the count of rear axles of the trailer.

6.6 Security

Vehicle classes 2 through 5 shall use IEEE Std 1609.2 security services and certificates as specified in SAE J2945/1. All other vehicle classes additionally include service specific permissions (SSPs) in their certificates indicating their permission to operate as the corresponding vehicle class. See Appendix B for the SSP design.

7. SYSTEM DESIGN

As noted in 6.5, conformance with SAE J2945/1 and/or SAE J3161/1 is necessary to conform to this document (SAE J2945/1B). Any necessary parameter settings or features not specified in this document are in SAE J2945/1 and/or SAE J3161/1.

7.1 Tutorial

Detailed information about dialogs may be found in NTCIP 8002 Annex B1 Section 4. Refer also to SAE J2945.

7.2 Specified Dialogs

This document uses a broadcast message dialog as described in SAE J2945.

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8. ASN.1 ENTITIES

The ASN.1 entities referred to in this document are defined in SAE J2735, and the tables in this section describe how they are used. Data Frames and Data Elements in addition to those used by SAE J2945/1 and SAE J3161/1 are used to provide information about trailers and motorcycles. The use of the BSM by implementations conformant with this document are specified in Table 12. The unique frames and elements needed to describe a pulling vehicle and its trailers and for motorcycles are listed in Tables 13 and 14, respectively. Changes that were made to SAE J2735 in support of this document are summarized in Table 15.

Table 12 - ASN.1 entities used in SAE J2945/1B

Title (SAE J2945/1B Clause)	Description
Message Encoding	Refer to SAE J2945/1.
MSG_MessageFrame	
Data Element: DE_DSRC_MessageID	
Message: MSG_BasicSafetyMessage Part I	
Message: MSG_BasicSafetyMessage Part II	
Data Frames	
Data Frame: DF_BSMcoreData	Refer to SAE J2735 definition. For vehicles with trailers, all contents of this Data Frame are for the pulling vehicle.
Data Frame: DF_VehicleSafetyExtensions	Refer to Part II of the Basic Safety Message as specified in SAE J2735. Used in conveying vehicle event flags, in this standard eventJackKnife is added.
Data Frame: DF_SpecialVehicleExtensions ***(Deprecate DF_TrailerData)***	Refer to SAE J3161/1 for use of DF_SpecialVehicleExtensions.
Data Frame: DF_TrailerData ***(deprecated)***	Not used.
Data Frame: DF_VehicleData ***(Additional Requirement)***	Part of DF_SupplementalVehicleExtensions. DF_VehicleData to include the following: ▪ trailerPresent ▪ DF_PivotPointDescription, used in providing the specifics of the connection point of the pulling vehicle. ▪ DF_Axles (new DF) ▪ rollAngle (motorcycles only) TrailerWeight to be removed from DF_VehicleData.
Data Frame: DF_ TrailersJ2945Slash1B ***(New)***	Added to DF_ SupplementalVehicleExtensions, a new Data Frame specifically used to provide information about one to eight trailers, provided in the order of connection to the pulling vehicle.
Data Frame: DF_TrailerUnitDescJ2945Slash1B ***(New)***	Contains the description of a single trailer, starting with the trailer connected directly to the pulling vehicle. Each trailer is described as a rectangular shape with certain width and length, with measurements of front offsets of its pivot point relative to the front edge of the rectangular body, and indication of front pivotability. The information about each trailer may also include information regarding: ▪ DF_PivotPointDescription (rearPivot) ▪ DF_Axles ▪ DE_J1939-71-Trailer Weight (Trailer weight) ▪ DE_VehicleHeight (Trailer height) ▪ DF_BumperHeights
Data Frame: DF_PivotPointDescription	Used in DF_ VehicleData to provide information related to the point of pivot on the pulling vehicle, and in DF_TrailerUnitDescJ2945Slash1B to provide the physical location of the points of pivot (front and rear) on a trailer. Includes indication of capability to pivot. For trailers, it provides pivot angle, measured between the center-line of this unit and the unit ahead that is pulling it.
Data Frame: DF_TrailerUnitDescription ***(deprecated)***	Not used.

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Title (SAE J2945/1B Clause)	Description
Data Frame: DF_Axles ***(New)***	The Data Frame contains counts of the axles of a vehicle or trailer: ▪ totalAxles ▪ frontAxles ▪ rearAxles
Data Elements	Note: The system includes the Data Elements from Part I and Part II of the Basic Safety Message.
Data Element: DE_Latitude	Per SAE J2945/1, provides the latitude of the center of the pulling vehicle regardless of any pulled segments.
Data Element: DE_Longitude	Per SAE J2945/1, provides the longitude of the center of the pulling vehicle regardless of any pulled segments.
Data Element: DE_Speed	Per SAE J2945/1, provides the speed of the pulling vehicle regardless of any pulled segments.
Data Element: DE_BasicVehicleClass	The BasicVehicleClass Data Element is mandatory for all non-light-duty vehicles.
Data Element: DE_VehicleWidth	Part of DF_VehicleSize in SAE J2735, it is used to provide the width of the vehicle. Also is used in DF_ TrailerUnitDescJ2945Slash1B to provide the width of the corresponding trailer. The width is the widest point of each unit of the vehicle with all factory-installed equipment.
Data Element: DE_VehicleLength	Part of DF_VehicleSize in SAE J2735, it is used to provide the length of the vehicle. Also, can be used in DF_ TrailerUnitDescJ2945Slash1B to provide the length of the corresponding trailer. Measured from the edge of the front bumper to the edge of the rear bumper of each unit it is used to define.
Data Element: DE_VehicleHeight	Per SAE J2735, as part of DF_ SupplementalVehicleExtensions, DE_VehicleHeight, provides the height of a pulling vehicle. DE_VehicleHeight is also used in DF_ TrailerUnitDescJ2945Slash1B to provide the height of each trailer. Measured from the ground to the highest surface of each unit.
Data Element: DE_TransmissionState	Per SAE J2735 the element used as part of BSMcoreDate in providing the movement status of a pulling vehicle.
Data Element: DE_Offset_B11	Per SAE J2735, it is used as part of the DF_PivotPointDescription in describing the physical location of the point of pivot relative to the edge of the unit the point is on. May be used for the vehicle or the trailer(s).
Data Element: DE_Angle	Referring to SAE J2735, the angle of pivot is provided relative to the unit directly ahead, pulling vehicle or trailer. In case of the pulling vehicle, the angle is not used.
Data Element: DE_PivotingAllowed	Referring to SAE J2735, Flag set to true when the described connection point allows pivoting to occur. (Part of DF_PivotPointDescription.)
Data Element: DE_VehicleEventFlags ***(Additional Requirement)***	eventJackKnife flag is to be added to the events of DE_VehicleEventFlags of SAE J2735.
Data Element: DE_TrailerMass ***(deprecated)***	Not used.

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Table 13 - Pulling vehicle and its trailers

Title (SAE J2945/1B Clause)	Description
Data Frame: DF_BSMcoreData	
Data Element: DE_Latitude	Per SAEJ2735. For rigid body vehicles, the same principles are applied. For vehicles with trailers, only the values corresponding to the pulling vehicle apply.
Data Element: DE_Longitude	
Data Element: DE_Speed	
Data Element: DE_TransmissionState	
Data Frame: DF_VehicleSize	
Data Element: DE_VehicleWidth	
Data Element: DE_VehicleLength	
Data Frame: DF_ SupplementalVehicleExtensions	
Data Element: DE_BasicVehicleClass	
Data Frame: DF_VehicleData	SAE J2735 definition is expanded to include trailerPresent, DF_PivotPointDescription, and DF_Axles.
Data Element: DE_VehicleHeight	
trailerPresent	Trailer present indication.
Data Frame: DF_Axles	
totalAxles	
frontAxles	
rearAxles	
Data Frame: DF_PivotPointDescription	Per SAE J2735 used to describe the point of pivot on the pulling vehicle.
Data Element: DE_Offset_B11	
Data Element: DE_Angle	Angle ignored in PivotPointDescription, set to 0
Data Element: DE_PivotingAllowed	
Data Frame: DF_ TrailersJ2945Slash1B	New DF encompassing information about one to eight trailers.
Data Frame: DF_TrailerUnitDescJ2945Slash1B	Details of each of the trailers are provided in elements within TrailerUnitDescription. Each TrailerUnitDescription provides the specifics of a single trailer. As many as eight sets (matching the quantity specified in TrailerUnitDescriptionList) may be included.
Data Element: DE_VehicleWidth	
Data Element: DE_VehicleLength	
Data Element: DE_VehicleHeight	
Data Element: DE_TrailerWeight	
Data Frame: DF_PivotPointDescription (front)	
Data Element: DE_Offset_B11	
Data Element: DE_Angle	
Data Element: DE_PivotingAllowed	
Data Frame: DF_PivotPointDescription (rear)	
Data Element: DE_Offset_B11	
Data Element: DE_Angle	
Data Element: DE_PivotingAllowed	
Data Frame DF_BumperHeights	
Data Element: DE_BumperHeight (front)	
Data Element: DE_BumperHeight (rear)	
Data Frame: DF_Axles	
totalAxles	
frontAxles	
rearAxles	
Data Frame: DF_VehicleSafetyExtensions	Used to convey the occurrence of a jackknife condition.
Data Element: DE_VehicleEventFlags	

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Table 14 - Motorcycles

Title (SAE J2945/1B Clause)	Description
Data Frame: DF_BSMcoreData	Per SAE J2735. For motorcycles, the same principles are applied. Motorcycle width is that of the widest part of the motorcycle, including handles.
Data Element: DE_Latitude	
Data Element: DE_Longitude	
Data Element: DE_Speed	
Data Element: DE_TransmissionState	
Data Frame: DF_VehicleSize	
Data Element: DE_VehicleWidth	
Data Element: DE_VehicleLength	
Data Frame: DF_ SupplementalVehicleExtensions	Motorcycle Specific Information. rollAngle added to DF_VehicleData.
Data Element: DE_BasicVehicleClass	
Data Frame: DF_VehicleData	
Data Element: DE_VehicleHeight	
rollAngle	

Table 15 - SAE J2735 change summary [informative]

	Change to SAE J2735 (*Additions* and ~~Deletions~~)
Data Frame: DF_SpecialVehicleExtensions	SpecialVehicleExtensions ::= SEQUENCE { ~~trailers TrailerData OPTIONAL~~
~~Data Frame: DF_TrailerData~~	Deprecate
~~Data Frame: DF_TrailerUnitDescription~~	Deprecate
Data Frame: DF_VehicleData	VehicleData ::= SEQUENCE { *trailerPresent OPTIONAL* *DF_Axles OPTIONAL* *rollAngle OPTIONAL* ~~TrailerWeight OPTIONAL~~ *DF_PivotPointDescription OPTIONAL*
Data Frame: DF_ TrailersJ2945Slash1B	*New Data Frame*
Data Frame: DF_TrailerUnitDescJ2945Slash1B	*New Data Frame*
Data Frame: DF_Axles	*New Data Frame* Axles ::= SEQUENCE { *totalAxles* *frontAxles OPTIONAL* *rearAxles OPTIONAL*
trailerPresent	Boolean
rollAngle	Integer
totalAxles	Integer
frontAxles	Integer
rearAxles	Integer
DE_VehicleEventFlags	*Add eventJackKnife*

9. NOTES

9.1 Revision Indicator

A change bar (I) located in the left margin is for the convenience of the user in locating areas where technical revisions, not editorial changes, have been made to the previous issue of this document. An (R) symbol to the left of the document title indicates a complete revision of the document, including technical revisions. Change bars and (R) are not used in original publications, nor in documents that contain editorial changes only.

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APPENDIX A - REQUIREMENTS TRACEABILITY MATRIX (RTM) TABLE

Table A1 - RTM table

FR ID	Functional Requirement	Dialog ID	Object ID	Object Name	Additional Specifications
6.4	Architectural Requirements				
6.4.1	Communications Requirements				
6.4.2	Positioning Requirements			DF_BSMcoreData DE_Latitude DE_Longitude	SAE J2735
6.5	Features			DE_BasicVehicleClass	SAE J2735
6.5.1	Non-Light-Duty Vehicle Class			DE_BasicVehicleClass	SAE J2735
6.5.2	Motorcycle Requirements			DF_SupplementalVehicleExtensions DE_BasicVehicleClass DF_VehicleData DE_VehicleHeight rollAngle	SAE J2735
6.5.2.1	Motorcycle Roll Angle			rollAngle	SAE J2735
6.5.2.2	Motorcycle Reported Height			DE_VehicleHeight	SAE J2735
6.5.3	Vehicles with Trailers Requirements				
6.5.3.1	Pulling Vehicle Attributes			DF_ SupplementalVehicleExtensions DF_PivotPointDescription	SAE J2735
6.5.3.1.1	Transmission State			DF_BSMcoreData DE_ TransmissionState	SAE J2735
6.5.3.1.2	Trailer Present Indication			DF_VehicleData trailerPresent	SAE J2735
6.5.3.1.3	Jackknife Flag			Added eventJackKnife to DE_VehicleEventFlags	SAE J2735
6.5.3.1.4	Pivot Ability			DF_ PivotPointDescription DE_PivotingAllowed	SAE J2735
6.5.3.1.5	Pivot Offset			DF_ PivotPointDescription DE_ Offset_B11	SAE J2735
6.5.3.1.6	Pulling Vehicle Total Number of Axles			DF_Axles totalAxles	SAE J2735
6.5.3.1.7	Pulling Vehicle Number of Front Axles			DF_Axles frontAxles	SAE J2735
6.5.3.1.8	Pulling Vehicle Number of Rear Axles			DF_Axles rearAxles	SAE J2735
6.5.3.1.9	Vehicles with Trailers Reported Height			DF_VehicleData DE_ VehicleHeight	SAE J2735
6.5.3.1.10	Vehicles with Trailers Reported Width			DF_BSMcoreData DE_VehicleWidth	SAE J2735
6.5.3.2	Information about Trailers			DF_TrailersJ2945Slash1B DF_TrailerUnitDescJ2945Slash1B	SAE J2735
6.5.3.2.1	Trailer Length			DF_TrailerUnitDescJ2945Slash1B DE_VehicleLength	SAE J2735
6.5.3.2.2	Trailer Front Pivot Ability			DF_TrailerUnitDescJ2945Slash1B DE_PivotingAllowed	SAE J2735
6.5.3.2.3	Trailer Rear Pivot Ability			DF_TrailerUnitDescJ2945Slash1B DE_PivotingAllowed	SAE J2735
6.5.3.2.4	Trailer Front Pivot Position Relative to Trailer's Front Bumper			DF_TrailerUnitDescJ2945Slash1B DE_ Offset_B11	SAE J2735
6.5.3.2.5	Trailer Rear Pivot Position Relative to Trailer's Rear Bumper			DF_TrailerUnitDescJ2945Slash1B DE_ Offset_B11	SAE J2735
6.5.3.2.6	Trailer Pivot Angle			DF_TrailerUnitDescJ2945Slash1B DE_ Angle	SAE J2735
6.5.3.2.7	Trailer Height			DF_TrailerUnitDescJ2945Slash1B DE_VehicleHeight	SAE J2735
6.5.3.2.8	Trailer Width			DF_TrailerUnitDescJ2945Slash1B DE_VehicleWidth	SAE J2735
6.5.3.2.9	Trailer Weight			DF_TrailerUnitDescJ2945Slash1B DE_VehicleWeight	SAE J2735

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6.5.3.2.10	Trailer Front Bumper Height			DF_TrailerUnitDescJ2945Slash1B DF_BumperHeights DE_BumperHeight (front)	SAE J2735
6.5.3.2.11	Trailer Rear Bumper Height			DF_TrailerUnitDescJ2945Slash1B DF_BumperHeights DE_BumperHeight (rear)	SAE J2735
6.5.3.2.12	Trailer Total Number of Axles			DF_TrailerUnitDescJ2945Slash1B DF_Axles totalAxles	SAE J2735
6.5.3.2.13	Trailer Number of Front Axles			DF_TrailerUnitDescJ2945Slash1B DF_Axles frontAxles	SAE J2735
6.5.3.2.14	Trailer Number of Rear Axles			DF_TrailerUnitDescJ2945Slash1B DF_Axles rearAxles	SAE J2735

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APPENDIX B - SSP DESIGN

The SSP design used in implementations conformant with this document is specified in this appendix.

In the case of public safety vehicles, this document uses the design of SAE J3161/1 and further extends that design per SAE J2945/5 to define SSPs for vehicles other than classes 2 through 5, while maintaining compatibility with SAE J3161/1. The additional SSPs used to support vehicle classes 1 and 6 through 13 are shown in Table B1.

An omitted SSP indicates only default behavior is allowed per SAE J2945/1 and/or SAE J3161/1. Note that all vehicle classes are permitted to include trailers when acting as a PV.

Table B1 - Entity activity groups: emergency vehicles

Entity Activity Group	Entity Activities	SSP Activity Bit
Public safety vehicle responding to an emergency (refer to SAE J3161/1)	Inclusion of DF_EmergencyDetails in the BSM. Inclusion of DE_BasicVehicleRole in the BSM, set to one of police (12), fire (13), and ambulance (14).	Bit 1 (bit 9 in the SSP)
FHWA vehicle class 1	Motorcycles, inclusion of rollAngle in the BSM.	Bit 2 (bit 10 in the SSP)
FHWA vehicle classes 6 and 7	Truck: single unit three- and four-axle trucks.	Bit 3 (bit 11 in the SSP)
FHWA vehicle classes 8 through 10	Truck: Single trailer.	Bit 4 (bit 12 in the SSP)
FHWA vehicle classes 11 through 13	Truck: Multi-trailer.	Bit 5 (bit 13 in the SSP)

The conformant SSP design that allows DF_EmergencyDetails and DE_BasicVehicleRole to be included in the BSM, and also allows for FHWA vehicle class to be indicated by fhwaVehicleClass within SupplementalVehicleExtensions, is as follows:

- Bits 1 through 8: Version Number. Set to one for this version of the specification (00000001).
- Bits 9 through 13: SSP activity bits per Table B1 (bits 14 through 16 are reserved).